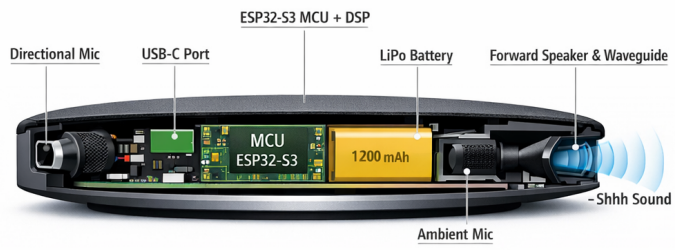
# Shusher

## The polite nudge you never had to give.

A palm-sized puck that listens for sustained loud speech, then fires a discreet, human-sounding "shhh" toward the offender. You aim it. It does the rest.



# The Problem

## Loud talkers and speakerphone abusers

Cafes, coworking spaces, trains -- someone always decides the whole room needs to hear their call. You tolerate it, or you confront a stranger. Neither feels good.

## Confrontation is awkward

Asking someone to lower their voice means making yourself the center of attention. Most people avoid it. The loud talker wins by default.

## No passive, low-effort solution exists

Noise-cancelling headphones help you, but they don't signal anything to the offender. There's no device that delivers a social nudge without making you the messenger.

**TARGET USERS**

**Remote workers | Students | Commuters**

**Library goers | Introverts | Anyone who values quiet**

# How It Works

## 1 Aim

**Point the puck at the noise source**

Place Shusher on your table and aim the front face toward the loud talker. Press the side button once. A brief vibration confirms it's active.

## 2 Detect

**Device listens and compares**

Two mics -- front (cardioid) and rear (omni) -- measure audio levels continuously. When the front mic detects sustained loud speech 10+ dB above the ambient baseline for 6+ seconds, the trigger fires.

## 3 Shush

**A natural "shhh" fires forward**

A pre-recorded human shush plays through a front-facing speaker with waveguide. Audible at the offender's table (2-5m), but blends with cafe noise for everyone else. Sounds human. Plausibly deniable.

## 4 Cool

**Wait, then repeat if needed**

After a shush, a 30-60 second cooldown prevents re-triggering. One polite nudge per incident. If they get loud again, another shush fires.

# Architecture

## Detection Chain

**Front mic --> Bandpass --> RMS --> Differential --> Trigger**

cardioid PDM  300-4kHz IIR  50ms window  front vs rear  sustained >6s

## Shush Chain

**Flash sample --> I2S --> MAX98357A --> Speaker + Waveguide**

## SUBSYSTEMS

### MCU (ESP32-S3)

Dual-core 240 MHz, audio DSP, BLE 5.0, 8MB flash

### Front mic (ICS-43434)

Cardioid via acoustic porting, PDM 16 kHz, 65 dB SNR

### Rear mic (ICS-43434)

Omnidirectional, ambient baseline reference

### Speaker (28mm + waveguide)

Forward-biased shush output, 80-85 dB @ 10cm

### Amp (MAX98357A)

I2S class-D, auto-shutdown when idle

### Power (1200 mAh LiPo)

USB-C charging, ~3 days per charge

## Designed to disappear

No visible LEDs during operation. No electronic sound artifacts. The shush sounds human -- breathy, variable, naturally timed. The offender wonders if someone nearby shushed them. Not a gadget.

# Plausible Deniability

The product works only if nobody knows it's there.

## Looks like a coaster

70mm diameter, 15mm tall, matte dark puck. No branding on top. Speaker slot disguised as a seam line. Mic port looks like a mold mark. Sits naturally on any cafe table.

## Sounds human

20+ pre-recorded shush variants across 5 styles: classic "shhh," throat clear, gentle "ahem," whispered "excuse me," and passive-aggressive sigh. Each firing picks a different sample. Breathy, slightly variable, naturally timed. No robotic quality, no beeps, no electronic artifacts.

## The discretion stack

- **No LEDs during operation:**
  A single LED under the device confirms power-on, then goes dark. Invisible on a table.
- **Forward-biased audio:**
  Waveguide provides ~4-6 dB front-to-side ratio. The target hears it; side-facing listeners don't.
- **Cooldown prevents harassment:**
  Max one shush per 30-60 seconds. A polite nudge, not a weapon. The offender self-corrects or doesn't.

# Constraints & BOM

**Battery life** — **~3 days**

1200 mAh LiPo, 4-hr daily active sessions

**Size** — **70mm x 15mm**

Coaster-sized puck, fits naturally on a table

**Shush range** — **2-5 meters**

Audible to the target, blends beyond 5m

**Trigger latency** — **~6 seconds**

Sustained loud speech, not brief bursts

**False positive rate** — **< 1 / hour**

Shushing the barista is a product killer

**Weight** — **~65g**

PCB 20g + battery 25g + enclosure 15g + speaker 5g

## BOM ESTIMATE (1k units)

| | |
|---|---|
| ESP32-S3-WROOM-1 (N8) | **$2.80** |
| Front MEMS mic (ICS-43434) | **$0.80** |
| Rear MEMS mic (ICS-43434) | **$0.80** |
| MAX98357A class-D amp | **$1.20** |
| 28mm dynamic speaker | **$0.60** |
| LRA haptic motor (8mm) | **$0.50** |
| LiPo battery (1200 mAh) | **$2.50** |
| MCP73831 charge IC + USB-C | **$0.80** |
| LDO + passives + RGB LED | **$0.50** |
| PCB (55mm round, 4-layer) | **$1.20** |
| Enclosure (2-piece + waveguide) | **$2.50** |
| Packaging + assembly + test | **$2.80** |
| **Total COGS** | **~$17.25 (5k: ~$13.50)** |

# Hardest Problems

## 1 Distinguishing shush-worthy noise from cafe ambiance

The front mic picks up everything in the aimed direction -- the loud talker, but also music, dish clatter, espresso machines. The classifier must isolate sustained loud speech above the ambient baseline and hold for seconds before triggering. Too sensitive: shushes the barista. Too conservative: never fires.

## 2 Making the shush sound convincingly human

If the shush sounds electronic or robotic, the product fails socially. It needs to sound like a real person -- breathy, slightly variable, naturally timed. Speaker coloration, waveguide resonances, and enclosure vibration could add artifacts. Multiple recordings with randomized selection on each firing.

## 3 Achieving useful directivity from a small speaker

A 70mm puck with a 28mm speaker and short waveguide won't produce a tight beam. Target: ~4-6 dB front-to-side ratio at 2-4 kHz (the "shhh" frequency range). Enough to make the shush louder for the target than for people at 90 degrees, but not a laser. Waveguide geometry and speaker placement are critical.

All three require physical prototyping. Acoustic porting, waveguide geometry, and sound recording quality can only be validated by building and testing in real environments.

# Gate Result & Next

**GATE: 74 PASS**

74 pass / 6 N/A / 8 fail

System description complete.

**8 items need prototype validation.**

## POWER

84 mA active | 2.3 mA standby | 1200 mAh battery

~3 days typical | USB-C charge | ~2.5 hr 0-100%

## KEY SPECS

70mm x 15mm puck | ~65g | ESP32-S3 | BLE 5.0

2 MEMS mics | Waveguide speaker | $50-70 retail

## KEY OPEN ITEMS

| | |
|---|---|
| M2 | Cardioid mic porting: 10-15 dB front-to-rear rejection? |
| M2 | Waveguide: >4 dB front-to-side ratio at 2-4 kHz? |
| M2 | Shush naturalness through 28mm speaker in plastic enclosure |
| M3 | False positive rate in 5+ real cafe environments |
| M2 | ESP32-S3 BLE + dual-mic audio coexistence under load |
| M2 | Sound recording session: 20+ shush variants, 5 styles |
| M1 | Market positioning: $50 tool vs. $30 fun gadget |
| M3 | DSP duty-cycle optimization for extended battery life |

Next step: breadboard prototype on an ESP32-S3 devkit with two MEMS mics and a small speaker. Test the front/rear differential in a real cafe. Record the shush samples. Validate before committing to PCB and enclosure tooling.